

BSc in IT: Specialising in Data Science

IT3021: Data Warehousing
and Business Intelligence

Lecture 02

Architecture & Components

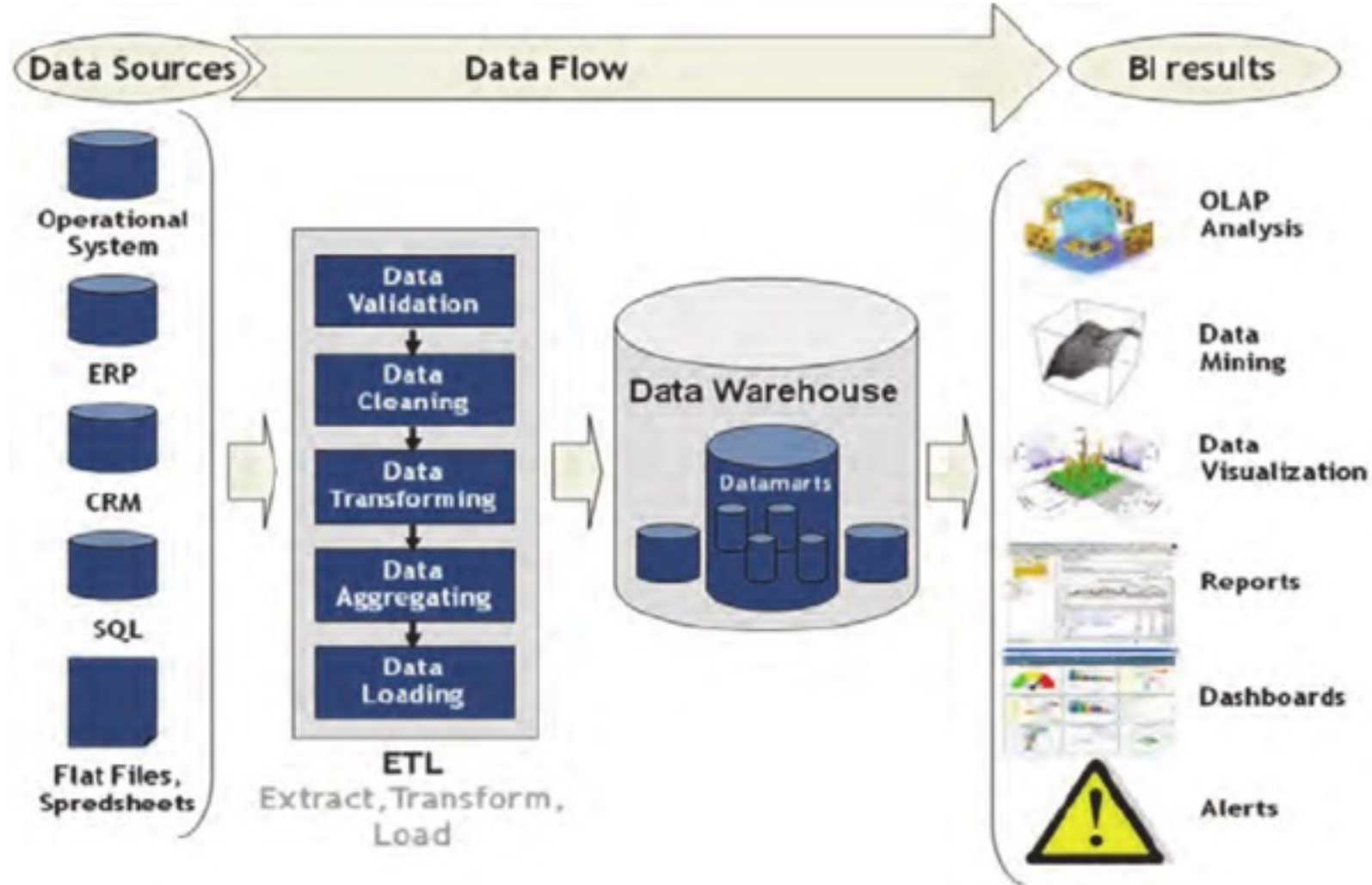
Content

- Main components of DW/BI
- Architectures
- Component details
- Required tools
- MS SQL Server components

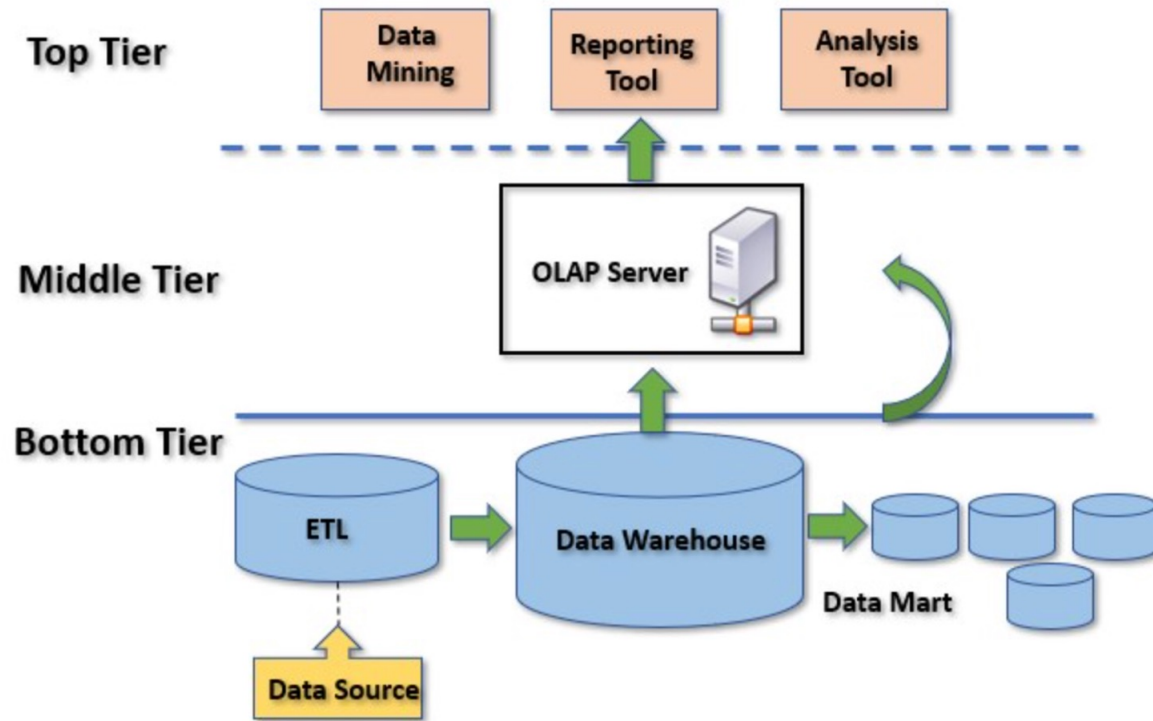
Main Components of a DW & BI Solution

- Data sources
- ETL/ELT
- Storage layer components
- Data consumption: BI/Advanced Analytics

A Simple Architecture



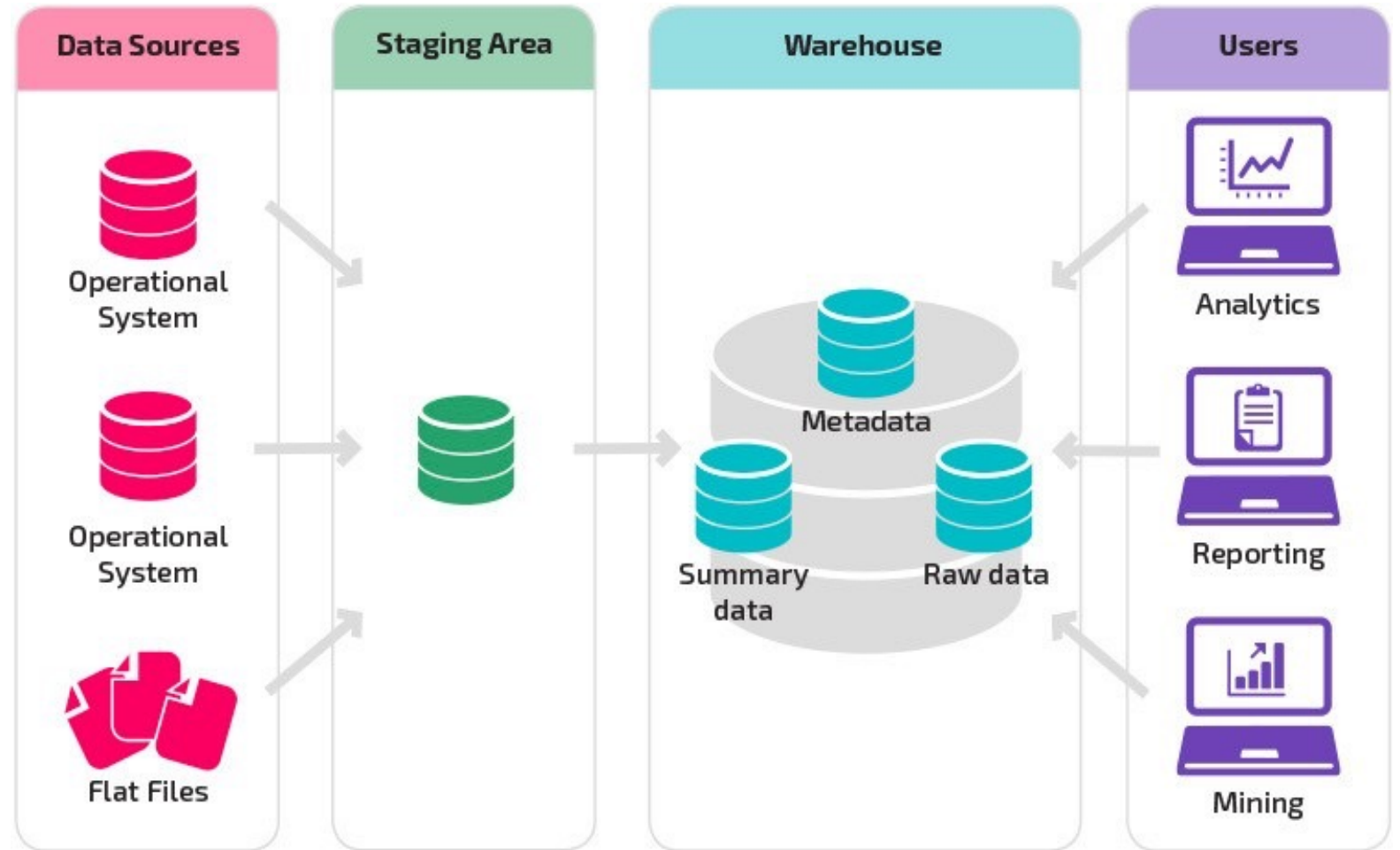
DW Architecture: 3 Tiered



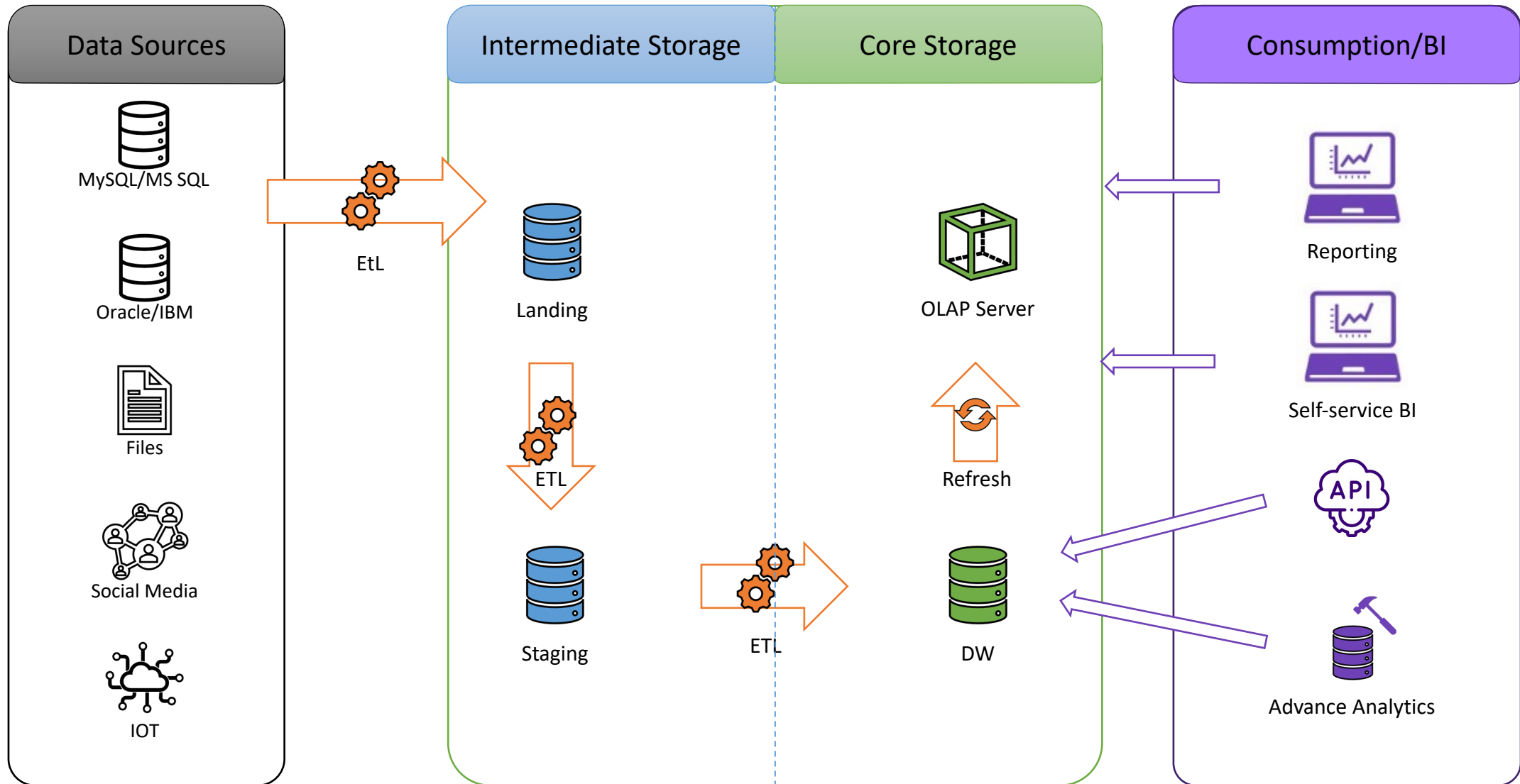
- **Top-Tier:**
 - The top tier is a front-end client layer.
 - Tools and APIs that you connect and get data out from the data warehouse.
 - It could be query tools, reporting tools, managed query tools, analysis tools and data mining tools.
- **Middle Tier:**
 - The middle tier is an OLAP server.
 - For a user, this application tier presents an abstracted view of the database.
 - Acts as a mediator between the end-user and the database.
- **Bottom Tier:**
 - The database of the data warehouse servers as the bottom tier.
 - It is usually a relational database system.
 - Data is cleansed, transformed, and loaded into this layer using back-end tools.

DW Architectures: with Staging Area

- Purpose:
 - Minimize dependency
 - Minimize impact



A Modern Architecture



Architectural Components: Data Sources

- Data sources can be:
 - Structured, semi-structured or unstructured
 - Disintegrated data
 - Internal or external data
 - Sometime, non-digitized
 - Examples:
 - Backend databases of operational systems (OLTP)
 - Files (text, csv, tsv, xml, etc.)
 - APIs
 - IOT devices
 - Web data (logs, click streams, etc.)
 - Social media data

Architectural Components: ETL

- Extraction: reading source data
 - Full vs. incremental (CDC)
 - Online vs. offline
 - Push vs. pull
- Transformation: preparing data to be inserted to the target model
 - Cleansing, integrating (matching & combining), de-duplicating, enriching, calculations, aggregations, etc.
- Loading: insertion of data to the target
 - Different types of tables (facts, dimensions, etc.)
 - Order of loading matters
 - Surrogate key generation
 - Hierarchy management

Architectural Components: Storage Layer

- Storage layer can be implemented in many different forms (next slide)
- Intermediate stages
 - Landing area
 - Staging area
- Data Modelling
 - Dimensional data model: facts and dimension tables
 - Multiple schema options: star, snowflake, etc.
 - Different levels of aggregations: row vs aggregate

Implementations of Storage Layer

- Operational Data Store:
 - ODS is a data store used when neither data warehouse nor OLTP systems support organizations reporting needs
 - ODS is refreshed in real time and hence used for real-time BI/analytical requirements
 - Often landing area is converted to ODS
- Enterprise Data Warehouse:
 - Centralized data warehouse which provides decision support services across the enterprise
 - It offers a unified approach for organizing and representing data
 - It also provides the ability to classify data according to the subject and give access according to those divisions
- Data Mart:
 - A subset of the data warehouse specially designed for a particular line of business, such as sales, distribution or finance
- Data Lake:
 - It is a centralized repository that allows you to store all your structured and unstructured data at any scale

Architectural Components: OLAP Engine

- Sometimes called '**Semantic Layer**': instead of an OLAP engine, aggregated tables can be used
- Storage modes:
 - ROLAP
 - MOLAP
 - HOLAP
- OLAP operations:
 - Slice
 - Dice
 - Roll-up
 - Drill-down

Architectural Components: Consumption/BI

- Operational reports
- Dashboards
- Ad-hoc analysis
- Self-service BI
- Advanced analytics
 - Data mining
 - Machine learning and predictive modelling
- Data for other operational applications
 - API

Architectural Components: Consumption/BI

- BI Applications
 - Web applications
 - Mobile applications
 - Self-service BI tools
 - BI Accessibility
 - Desktop/Laptop
 - Mobile devices
- Other
 - Data mining/modelling tools
 - APIs
 - Direct data access (a source for other systems)

Need for DW/BI Tools

- Implementation of storage layer components
 - ODS/EDW/data mart/data lake
 - Staging/landing areas
- Implementation and automation of ETL processes
 - Data cleansing tools
 - Data validation tools
- Implementation of analytical storage layer
- Implementation of BI components and other methods of consumption

MS SQL Server Components

- MS SQL Server
 - Database Engine
 - SSIS
 - SSAS
 - SSRS
- SQL Server Management Studio (client access tool)
- SSDT (development IDE)
- Report Builder (development IDE)
- DQS
- Power BI
- MS Excel

Use of MS BI Components in DW

- Implementation storage layer components:
 - Database engine
 - SQL Server Management Studio – client access tool
- Implementation and automation of ETL processes:
 - SSDT – IDE for ETL flow developments
 - SSIS – ETL engine
 - SQL Server Agent – Process automation
 - SQL Server Management Studio – SSIS repository access

Use of MS BI Components in DW

- Implementation of analytical storage layer:
 - SSDT – IDE for Cube developments
 - SSAS – Analytical processing and data mining engine
 - SQL Server Management Studio – SSAS repository access
- Implementation of BI components:
 - SSDT – IDE for report developments
 - Report Builder - IDE for static report developments
 - Mobile Report Publisher – IDE for mobile report developments
 - Report Server Configuration Manager – Integration and administration of SSRS web application
 - SSRS – Server based report generation, administration and hosting platform
 - SQL Server Management Studio – SSRS repository/meta data access
 - Power BI – Cloud based visualization platform/Self-service BI
 - MS Excel